



瑞奇科技
READCELLS SCIENTIFIC



CELLImage

Automated live cell imaging system

Observation and analysis of cells in the incubator

No interference, Long-term, High-throughput

◆ CELLImage Automated live cell imaging system

CELLImage Automated live cell imaging system can be placed in a cell culture incubator for use, without being affected by weak acids, high temperatures, and high humidity inside the incubator, reducing the damage of traditional observation methods to cells. It is fully automatic, long-term, no interference, high-throughput, and can dynamically capture and analyze precise cell images in real time, suitable for observation in all cell experiments. The device is equipped with an optical system, imaging system, hidden stage system, and software controlled analysis system. By setting up experimental programs, the incubator automatically observes cells at regular intervals and takes photos for storage, intelligently analyzes and identifies cells, and has multiple image data analysis functions.



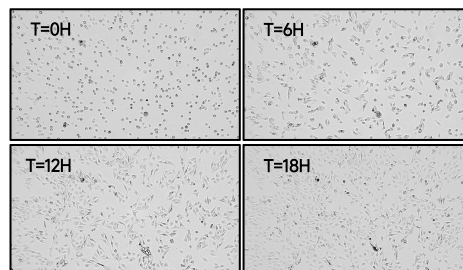
CELLImage Nano

CELLImage Nano: Automatically monitor the dynamic changes of cells at unit points (single well positions) in the incubator and analyze cell images in real-time

Features: Compact and convenient, suitable for different sizes of incubators, dynamic imaging with bright field and monochromatic fluorescence



Tracking and monitoring changes in cell growth inside the incubator

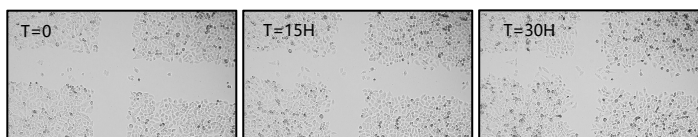


CELLImage Mini

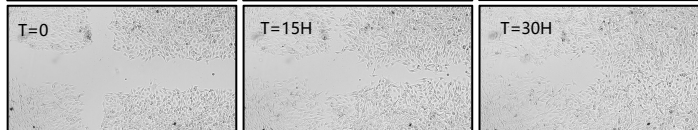
CELLImage Mini: Automatically monitor the dynamic changes of cells at multiple locations (porous sites) inside the incubator, record and analyze the differences in changes of cells at different locations

Features: Multi point imaging, unlocking cell control experiments, supporting automatic monitoring of high-throughput samples in 384 well plates, bright field, blue fluorescence, green fluorescence, red fluorescence

CG



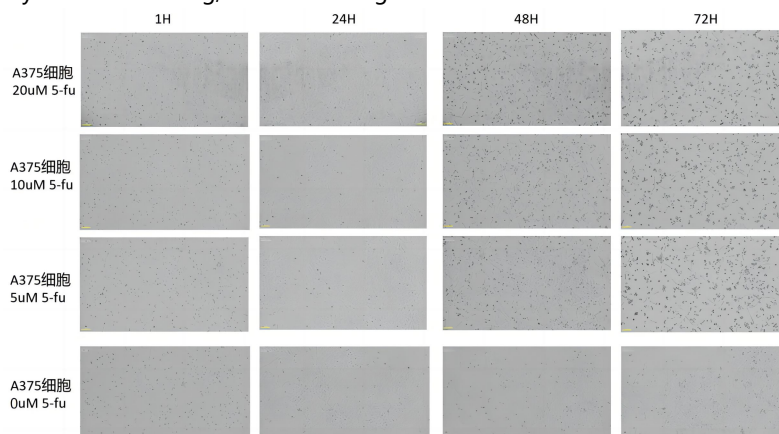
EG



CELLImage Mini Pro

CELLImage Mini Pro: Automatically monitor the dynamic changes of cells in the double plate position and multi position (porous position) inside the incubator, record and analyze the differences in changes of cells at different points of the two plate positions

Features: Two sample plate positions, double high flux bright field+fluorescence dynamic monitoring, standard blue green red 3-channel fluorescence

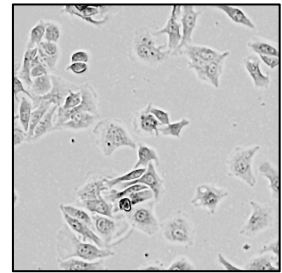
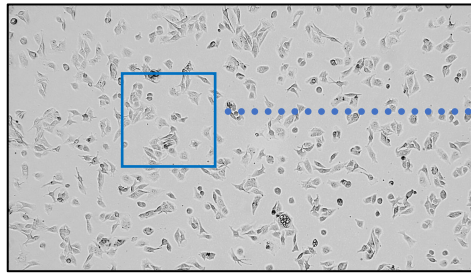


High throughput cellular drug screening experiment

CELLImage: Multiple imaging channels including bright field and fluorescence

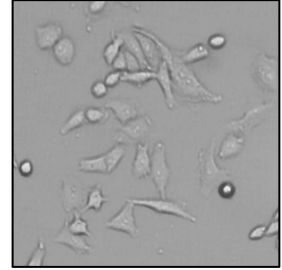
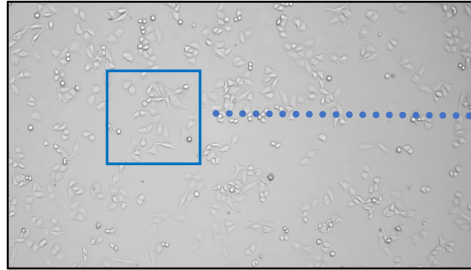
Bright field imaging

Bright Field 4K Imaging: CELLImage uses an 8-megapixel physical pixel back illuminated high-resolution CMOS to achieve clear imaging at 4K resolution and restore the true morphological changes of cells.

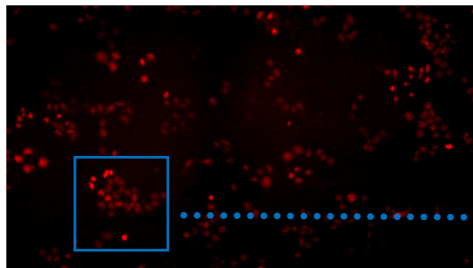


Bright field 4K imaging

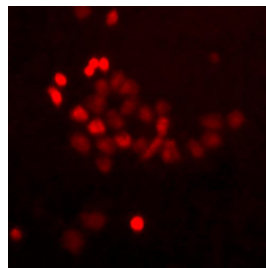
Mingchang Digital Stereoscopic Imaging: Our self-developed digital stereoscopic imaging technology enables the acquisition of more three-dimensional cell images through Mingchang channels.



Bright field digital stereo imaging

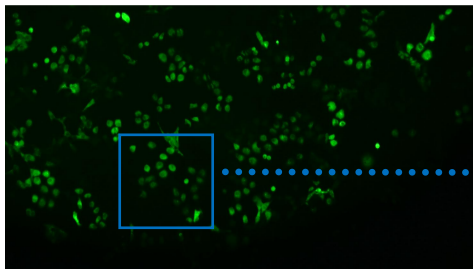


mcherry

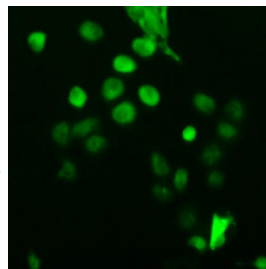


Fluorescence imaging

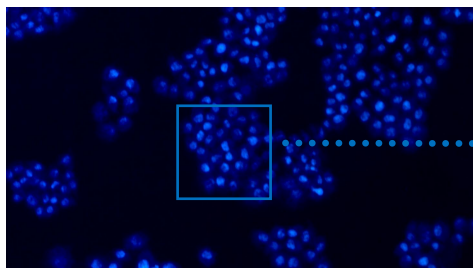
The 8 million resolution CMOS used by CELLImage not only supports bright field 4K resolution imaging, but also is compatible with high-resolution fluorescence imaging, restoring the true fluorescence changes of cells.



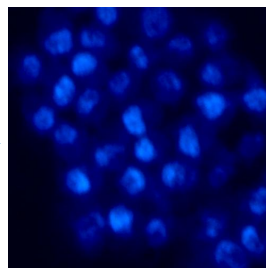
GFP



CELLImage supports fluorescence imaging in three channels: red, green, and blue. It adopts a highly specific fluorescence module design, which has the advantages of high sensitivity, strong specificity, and multiple labeling. It can not only meet the imaging needs of fluorescence staining, but also achieve dynamic monitoring of fluorescence with appropriate fluorescent dyes.



DAPI



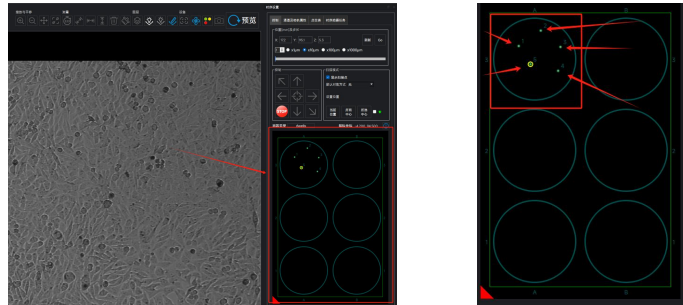
Fluorescence wavelength reference

Red: EM 605/55
Green: EM 535/40
Blue: EM 460/50

◆ CELLImage: Multiple imaging modes

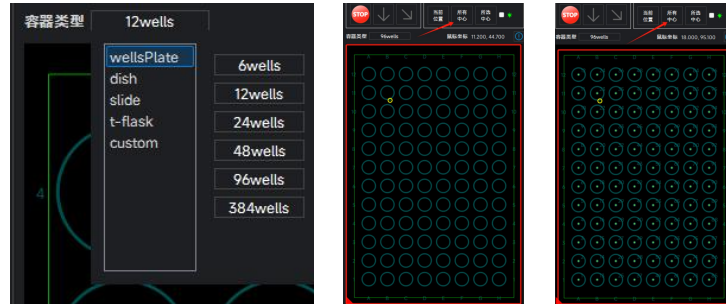
Free field imaging mode

CELLImage can freely view any position of the 117 x 77mm super large imaging area (different hole positions, hole edges, hole centers, any position of holes), without limiting the number of automatic imaging sites, and the ultra-high degree of freedom can meet different experimental needs.



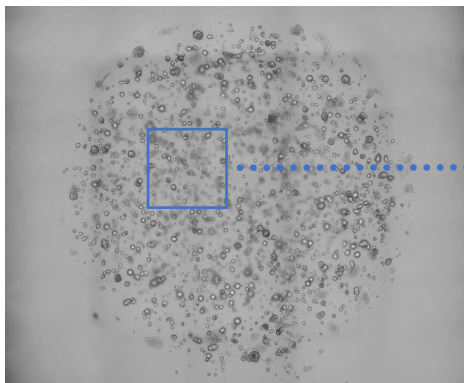
Fixed field of view

CELLImage has built-in data of various cell culture containers, which can quickly select the center point of different containers (for example, the center point of 96 well plate /384 well plate can be set with one key) and automatically monitor the fixed field of view.

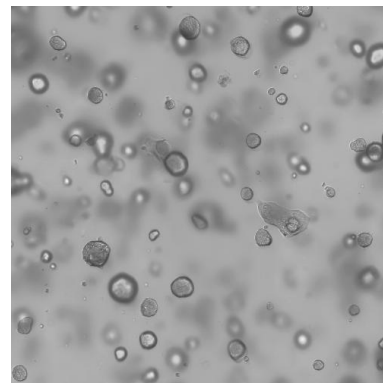


Large image scanning

CELLImage supports the free setting of the site and range of large image scanning. The self-developed traceless image stitching algorithm ensures that the cells in the stitched position have no dislocation and overlap, and perfectly restores the real change image of the cells in the super large field after stitching.



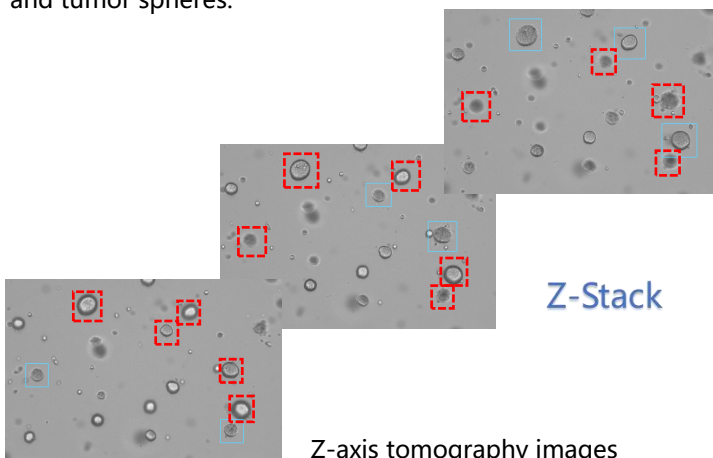
35 image mosaic organoid map



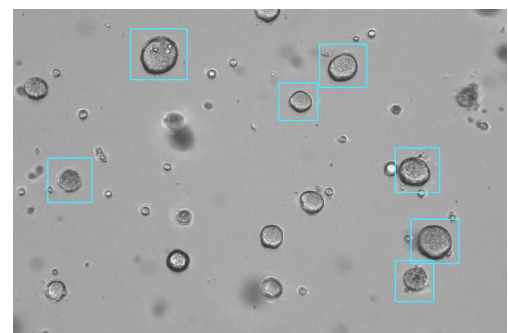
Mosaic position effect of 4 images

Z-stack

CELLImage supports free setting of z-axis slice scanning interval step size, unlimited scanning range, and self-developed Z-stack algorithm to meet the observation needs of 3D cell like samples such as organoids and tumor spheres.



Z-axis tomography images of lung tumor organoids

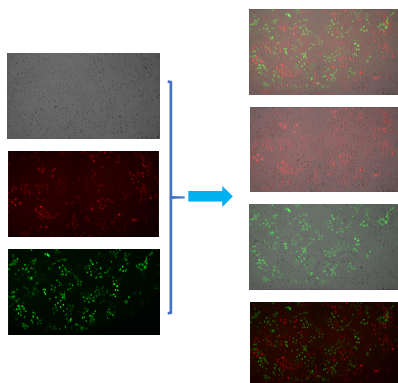


Z-stack overlay image of lung tumor organoids

◆ CELLImage: Multiple analysis modules

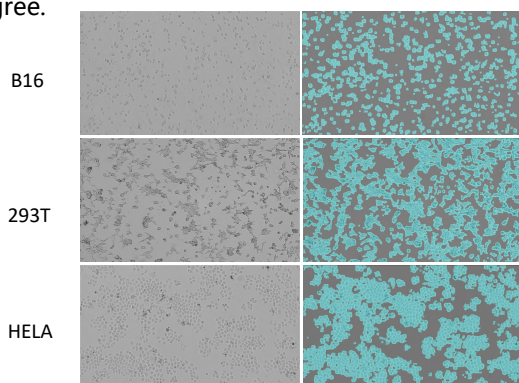
Multichannel merge

CELLImage supports multi-channel simultaneous imaging, and the cell images of different channels are automatically merged, and can be output with any layer. The image format is jpg/png/tiff



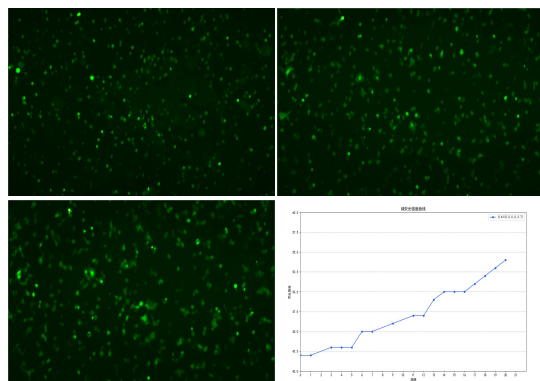
AI cell recognition

CELLImage self-developed intelligent cell recognition algorithm, with high degree of freedom parameter adjustment, accurately identifies cells of different shapes, breaks subjective constraints, and provides the experimenter with the standard of cell fusion degree.



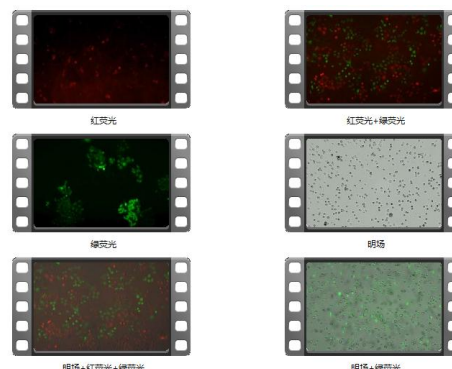
Fluorescence intensity

CELLImage self-developed intelligent cell fluorescence recognition algorithm can accurately recognize cell fluorescence, automatically calculate fluorescence intensity, generate fluorescence intensity curve, and quickly obtain experimental results.



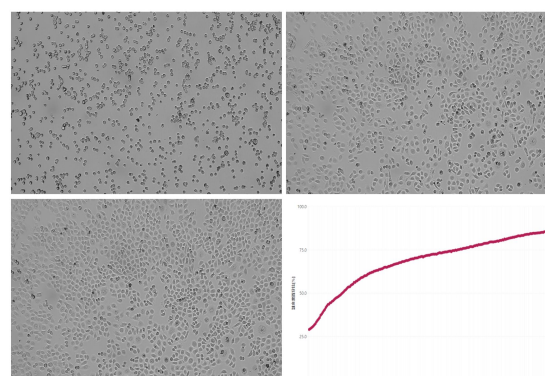
Delayed dynamic video

CELLImage long-term dynamic monitoring of cell growth changes, self-developed software automatically generates cell growth delay dynamic video, supporting bright field, fluorescence and multi-channel superimposed dynamic video.



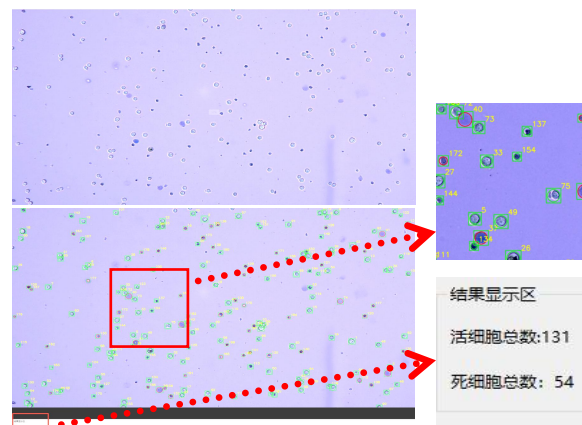
Cell growth curve

CELLImage self-developed intelligent cell recognition algorithm accurately identifies cells of different shapes, automatically calculates cell fusion, generates cell growth curves, and quickly obtains experimental results.



Trypan blue cell count

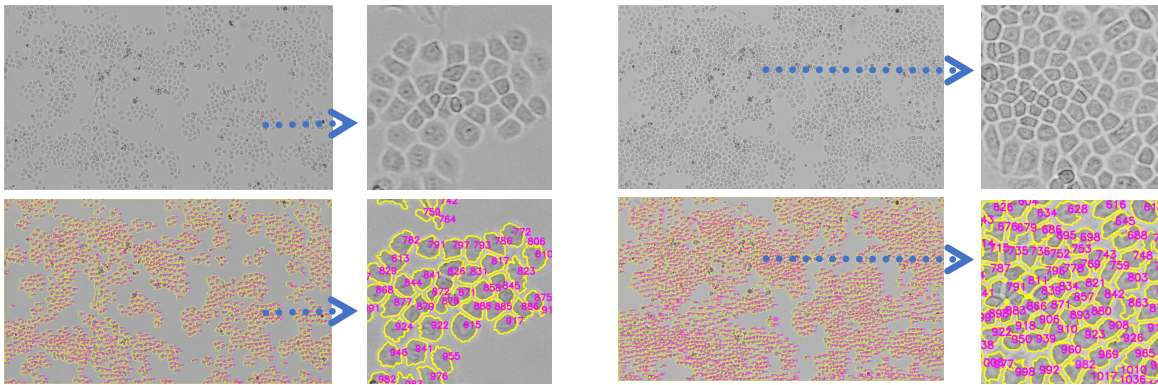
CELLImage self-developed intelligent cell recognition algorithm accurately identifies cells, accurately judges dead cells stained with trypan blue, and counts the number of dead and living cells respectively.



◆ CELLImage: Multiple analysis modules

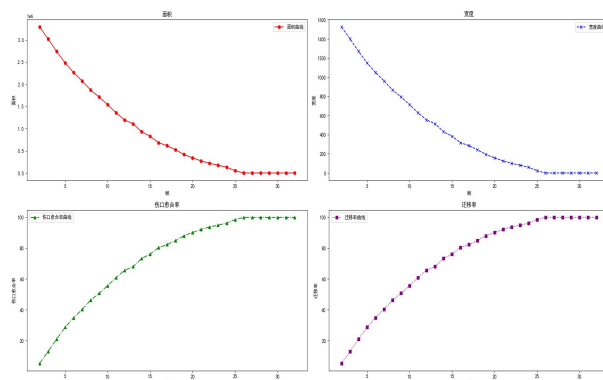
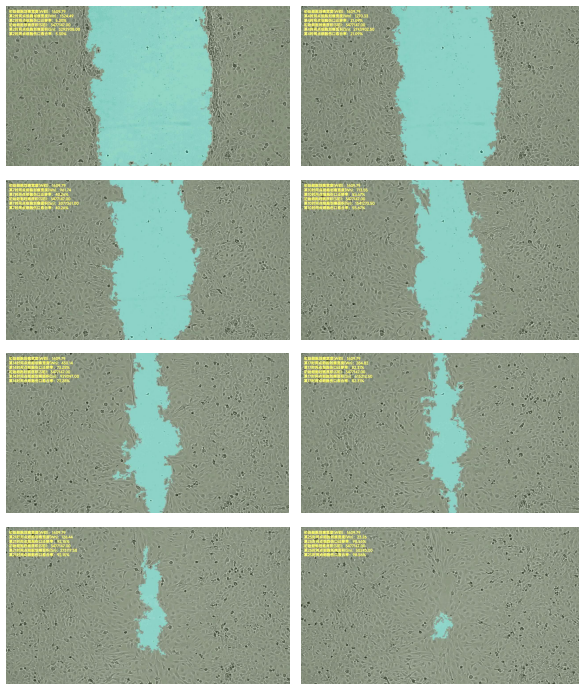
Label free counting of living cells

CELLImage self-developed intelligent label free live cell recognition algorithm can accurately identify and segment scattered and clustered live cells, and accurately count suspended and adherent cells.



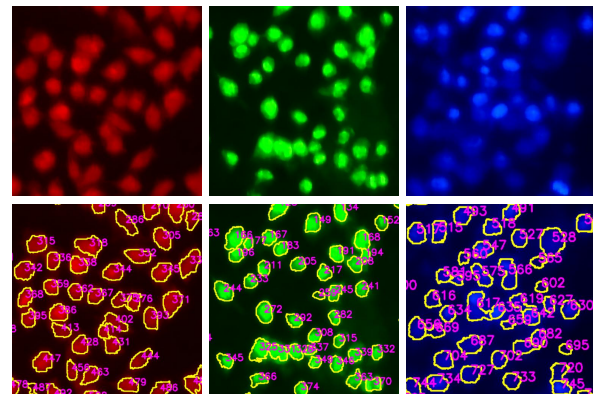
Cell scratch assay analysis

CELLImage self-developed intelligent cell scratch experimental analysis algorithm accurately identifies the cell scratch area, automatically calculates the average scratch width and scratch area, and outputs data result charts such as cell scratch wound healing rate and cell migration rate.



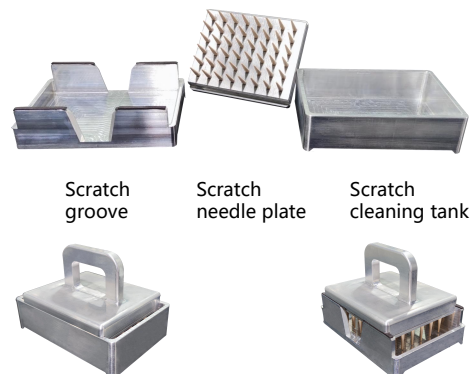
Fluorescent cell count

CELLImage self-developed intelligent fluorescent cell recognition algorithm accurately identifies and segments scattered and clustered fluorescent cells, and counts fluorescent cells in red, green and blue channels.



Scratch device

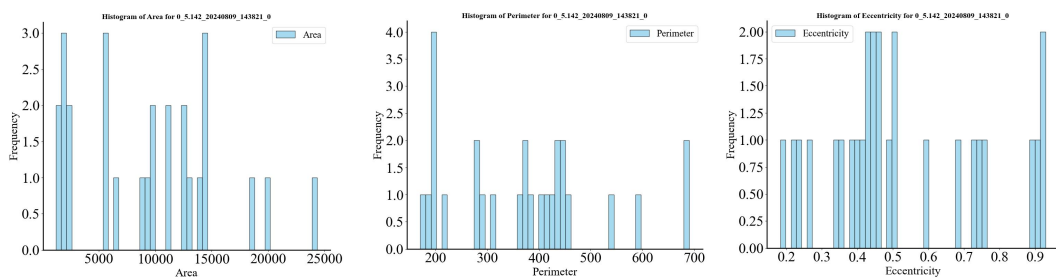
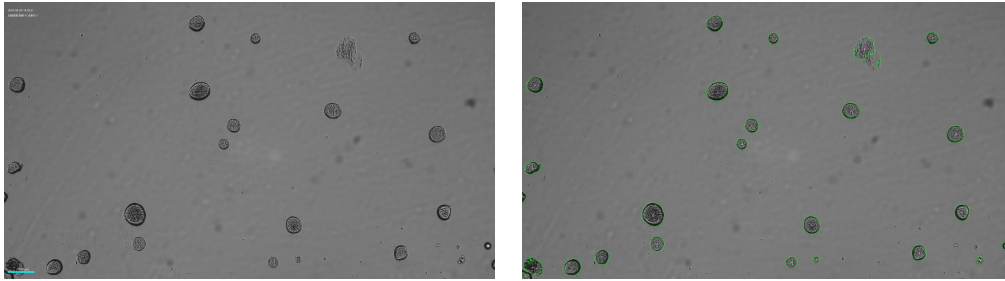
The self-developed scratch device can quickly scratch cells on 48 well plates, 96 well plates and other cell culture containers, and the scratches do not damage the culture container. It can be used with the full-automatic live cell imager to set the scratch dynamic monitoring program with one button, reduce manpower, improve work efficiency, and quickly obtain experimental results.



◆ CELLImage: Multiple analysis modules

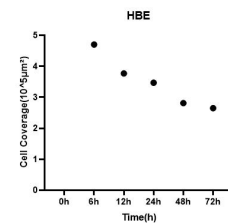
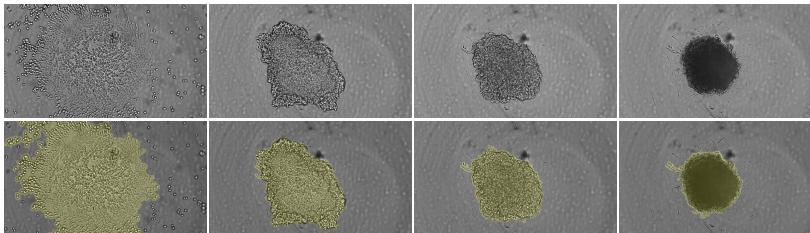
Organoid analysis

CELLImage self-developed intelligent organoid analysis algorithm can accurately identify organoids, automatically calculate the number, area, perimeter and other index data of organoids, and output the organoid growth trend chart.



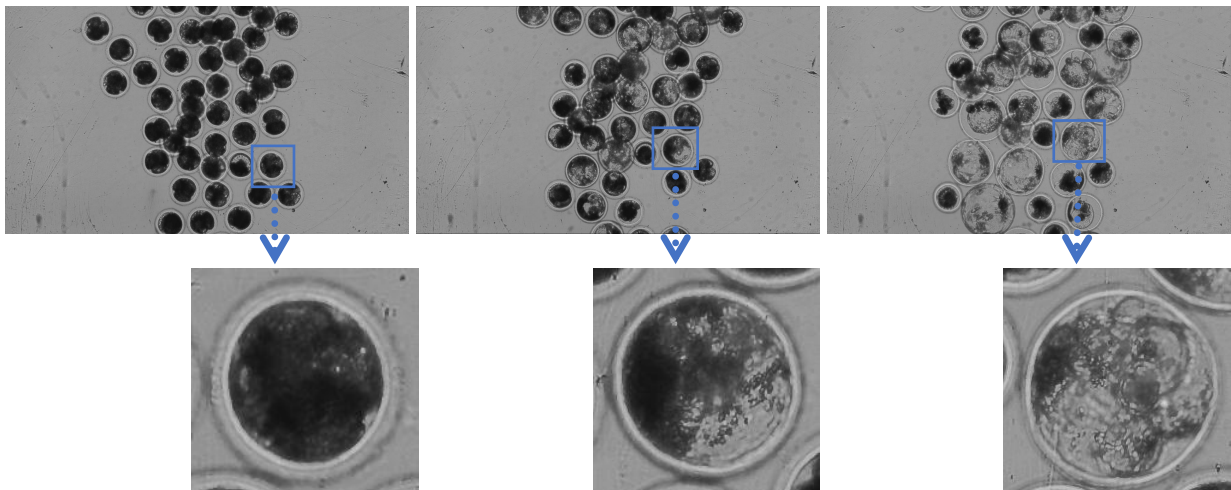
Tumorsphere analysis

CELLImage self-developed intelligent tumor sphere analysis algorithm accurately calculates the tumor sphere area, perimeter and other index data, and outputs the tumor sphere growth trend chart.



Dynamic monitoring of animal embryos

CELLImage self-developed animal embryo dynamic monitoring algorithm to track and monitor the development and growth of embryos (embryo cycle recognition and embryo intelligent scoring algorithm are under development).



Model parameters And Ordering information

Model	CELLImage Nano	CELLImage Mini	CELLImage Mini Pro
Functions	Automatic dynamic imaging of unit point in incubator	Multi site automatic dynamic imaging in incubator	Automatic dynamic imaging of two plates in incubator
Bright field imaging	LED light source, white and green light are optional		
Fluorescence imaging	Single channel fluorescence imaging	single channel green fluorescence imaging(optional) Blue: EM 460/50 Green: EM 535/40 Red: EM 605/55	Blue、Green、Red three channel fluorescence imaging Blue: EM 460/50 Green: EM 535/40 Red: EM 605/55
Magnification	10X		
Focus	Electric z-axis (Z-axis travel 10mm, movement accuracy 1um) + support automatic / manual focusing		
Stage	nothing	Electric hidden high-precision stage, XY axis movement accuracy 1um, support automatic / manual positioning	
Sample bit	Unit point	Single board, multi site	Double position, multi position
Camera	8MP CMOS		
Resolution ratio	4K imaging, 3840 x 2160 pixels		
Work environment	Temperature: 5 - 40 °C, humidity: 5 - 95%		
Connection mode	Single network cable connection to realize power supply, control and data transmission		
Application scenarios	Observation imaging, dynamic imaging, automatic imaging of cell experiments		
Adapter container	Transparent cell culture container ≤ 70mm (perforated plate, culture bottle, culture dish, glass slide, microfluidic, organoid chip)		

Model	Order	Ordering information description
CELLImage Nano	CINB1000	• one set of attached accessories; • b: open field; S: Monofluorescence; D: Dual fluorescence; T: Tri fluorescence; • green: 01; Red: 02; Blue: 03; Red + Green: 04; Green + Blue: 05; Red + Blue: 06
CELLImage Nano	CINS1001 (/02)	
CELLImage Mini	CIMB1000	
CELLImage Mini	CIMS1001 (/02/03)	
	CIMD1004 (/05/06)	
	CIMT1000	
CELLImage Mini Pro	CIPT1000	
Note: other special requirements such as 20x model and various non-standard adapters can be customized		

Company profile

Chongqing Readcells Scientific Co., Ltd., China is a national high-tech enterprise and a technological and innovative enterprise in Chongqing. In response to the call of the state, rich technology has made independent innovation, focused on live cell imaging and analysis technology, relied on the industrial cooperation of domestic colleges and universities, attracted domestic excellent technical teams, developed biomedical instruments with independent intellectual property rights, and provided high-end scientific research instruments with high price performance for domestic scientific researchers. At present, there are more than 10 patents related to live cell imaging, nearly 10 computer software works, and dozens of patents and soft works at the stage of application and approval, aiming to become an international leading enterprise in the field of live cell imaging.



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Scan the code to follow the public / video number and view the cell dynamic image